

K-means Clustering for Authorship Stylometry

Given a set of articles written by K authors, how well can we recover the author groups using only writing style? Perhaps the most natural algorithm to study this question is K-means.

1 Clustering 10 Authors

Dataset and preprocessing. We use the *LessWrong 10 Authors* dataset: 10 authors, 40 articles each, evenly split across two periods (20 from ≤ 2022 , 20 from ≥ 2024). To keep article length from masking the stylistic signal, we truncate every article to its first **1,526 words**, the global minimum length across the entire cleaned corpus, and discard anything past the cutoff.

Feature extraction. We extract **107** known stylometric features per article using `NLTK` for tokenisation and POS-tagging (with a regex fallback when NLTK is unavailable). The features fall into seven groups:

- **Word and sentence length** statistics (mean, standard deviation, median, fractions of 1–6 character words, etc.)
- **Vocabulary richness**: type-token ratio, hapax ratio, Yule K , Simpson D , Brun t W , Honor  R
- **Function-word frequencies**: usage rate of each of the top-50 English function words (“the”, “and”, “I”, “you”, ...)
- **Stylistic word categories**: rates for hedges, amplifiers, discourse markers, and conjunctions
- **Punctuation rates**: commas, periods, semicolons, parentheses, etc.
- **Coarse POS-tag distributions**: noun, verb, adjective, adverb, pronoun, determiner, preposition, conjunction, modal fractions
- **Character-level ratios**: uppercase fraction, vowel frequencies

Features with zero variance across the corpus are dropped; the rest are z-scored with `sklearn`’s `StandardScaler` so every feature has mean 0 and standard deviation 1 before clustering.

Clustering. We run `sklearn.cluster.KMeans` with $K = 10$, `init="k-means++"`, and `n_init=10`: the algorithm makes ten independent attempts and returns the one with the lowest **within-cluster sum of squares**, the *distortion*

$$L = \sum_{k=1}^K \sum_{\mathbf{x} \in C_k} \|\mathbf{x} - \boldsymbol{\mu}_k\|^2,$$

where C_k is the set of articles assigned to cluster k and $\boldsymbol{\mu}_k$ is its centroid.

Measuring performance: the Adjusted Rand Index. A natural measure of clustering quality compares the predicted cluster assignments against the true author labels by examining every *pair* of articles. For a dataset of n articles there are $\binom{n}{2}$ such pairs. For each pair, the clustering either **agrees** with the ground truth (both labellings put the two articles in the same group, or both put them in different groups) or it **disagrees**.

The **Rand Index (RI)** is simply the fraction of pairs on which the clustering agrees:

$$RI = \frac{\text{number of agreeing pairs}}{\binom{n}{2}}.$$

The problem is that even a random clustering achieves a high RI. With ten authors and 400 articles most pairs are from different authors; a random partition will still separate them most of the time, racking up agreements for free. The **Adjusted Rand Index (ARI)** subtracts the agreement expected from a random partition, rescaling so that random chance scores exactly 0:

- **ARI = 1**: perfect recovery of the true author groups.
- **ARI $\approx$ 0**: no better than random chance.
- **ARI < 0**: systematically worse than random.

ARI is also invariant to how cluster IDs are numbered, which matters because K-means returns arbitrary cluster IDs that need not match any author ordering.

Results. The 400-article input matrix is 400×107 after feature extraction.

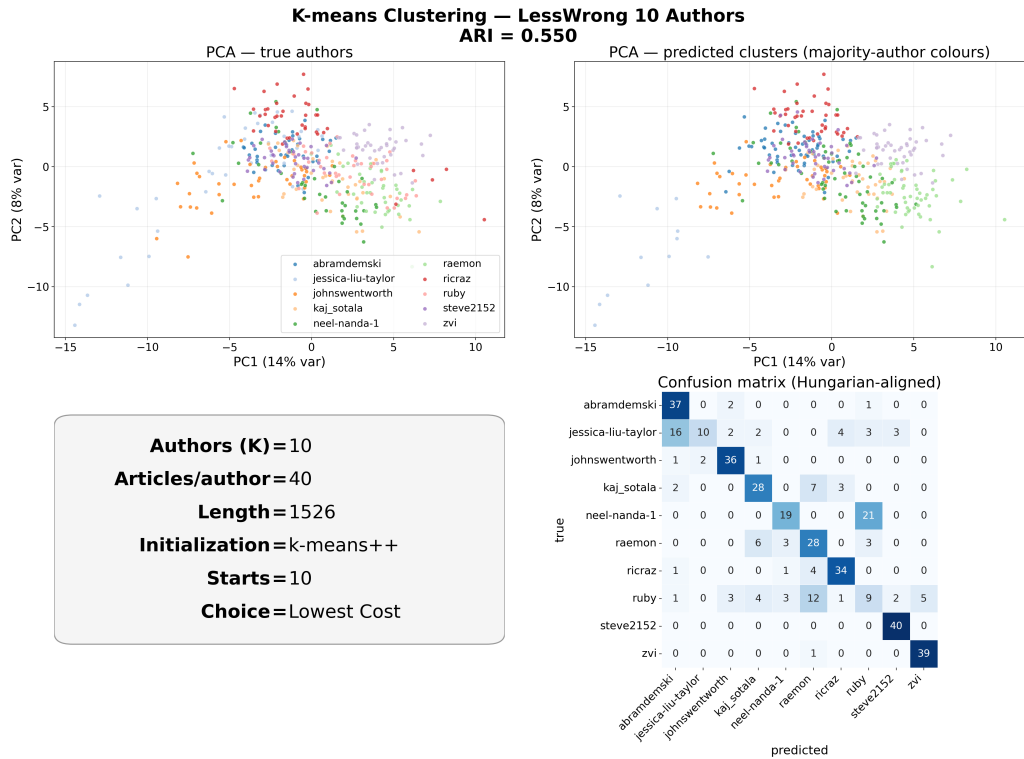


Figure 1: K-means on the LessWrong 10 Authors dataset, $K = 10$. *Top left*: articles projected onto the two PCA directions of largest variance (`sklearn.decomposition.PCA` applied to the centered, z-scored matrix), coloured by true author. *Top right*: same projection, coloured by predicted cluster (each cluster takes the colour of its majority author). *Bottom right*: confusion matrix after a Hungarian one-to-one alignment (`scipy.optimize.linear_sum_assignment`) between predicted clusters and true authors.

The clustering achieves **ARI = 0.55**, which is fairly impressive given that we are trying to separate ten groups

simultaneously. The confusion matrix shows a prominent diagonal: most authors land mostly in their own cluster.

Despite this strong result, a question remains: if we were to run the exact same experiment again, would we still see a similar ARI? K-means is a randomized algorithm. To cluster the articles, it must first place K initial centroids somewhere in the feature space. Different starting positions lead to different final clusters, and the algorithm can get stuck in different local minima of L . **k-means++** is a smarter initialization strategy than placing centroids uniformly at random: it places each subsequent centroid with probability proportional to its squared distance from the nearest centroid already chosen, biasing the starting configuration toward spread-out positions and improving convergence. But k-means++ is still random.

The randomness is controlled by a single integer called the *master seed* (`random_state` in sklearn). Given a seed, sklearn generates a fully deterministic sequence of random numbers. For each of the `n_init=10` attempts, k-means++ draws the next values from this sequence to select K starting centroids. The ten attempts therefore use different starting centroids (drawn sequentially from the same seed’s stream), each converges independently to some local minimum, and each records its own distortion L_i . sklearn returns the attempt $i^* = \arg \min_i L_i$ with the lowest distortion.

The key issue is that all ten attempts are tied to one seed. Change the seed, and sklearn generates a different random-number sequence, produces a different pool of ten starting configurations, and may return a different “best” clustering with a different ARI. With `random_state=42` we obtained $\text{ARI} = 0.55$, but without running other seeds we cannot tell whether this is a typical outcome or a lucky one. To measure the variability, we treat each seed as an independent trial: for ten seed values (42, 100, 200, $\dots$, 900) we run the same experiment and record the ARI of the lowest-distortion attempt. Figure 2 illustrates the procedure.

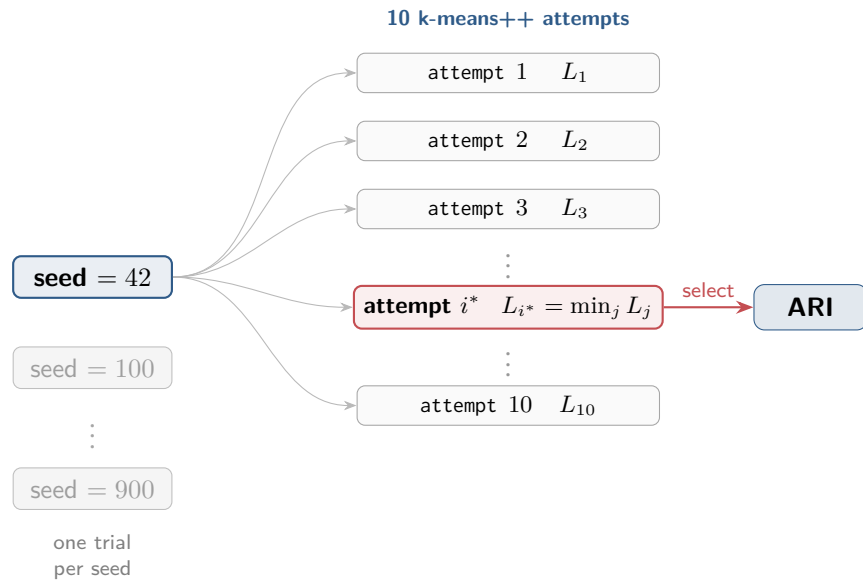


Figure 2: Procedure for one stability trial. The master seed generates a deterministic random-number sequence; k-means++ draws from it to produce ten independent starting configurations. Each attempt converges to a local minimum of L , and the one with the lowest distortion L_{i^*} is selected and scored with ARI. Running ten different seeds gives ten independent ARI measurements.

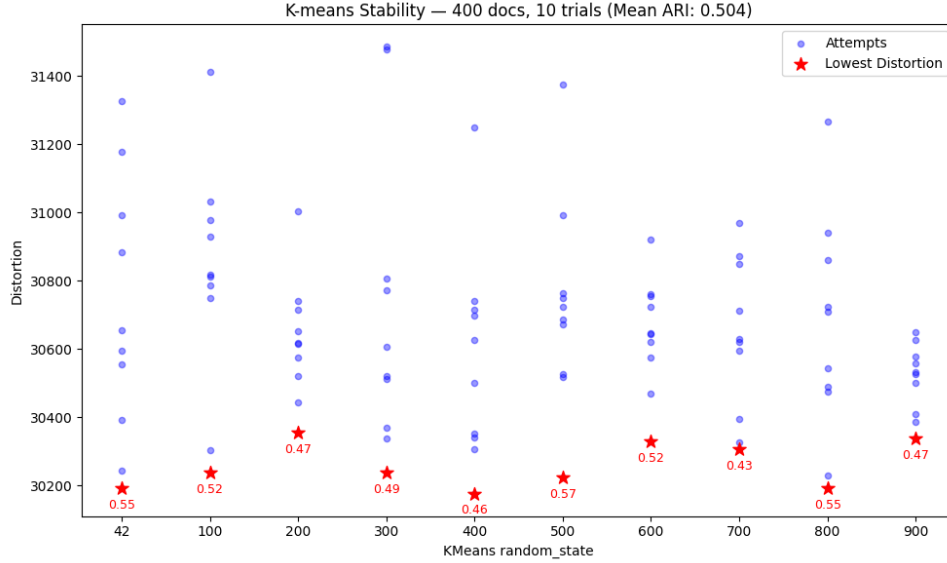


Figure 3: ARI of the selected attempt (lowest L) across ten master seeds. Blue dots show the distortions of all ten attempts within each trial; the red star marks the one sklearn returns. The annotated value is that run’s ARI.

Across the ten trials the mean ARI is **0.504**, ranging from **0.43** to **0.57**. Our initial result of 0.55 sits on the luckier side of the distribution, but it is close to the mean. The clustering quality we reported is real, not a fluke of a single seed.

Which features are doing the work? For every feature we compute the ANOVA F-ratio (between-cluster variance divided by within-cluster variance) on the predicted clusters using our `f_ratio` function in `src/clustering.py`. A high F-ratio means the feature separates clusters more than it varies inside them.

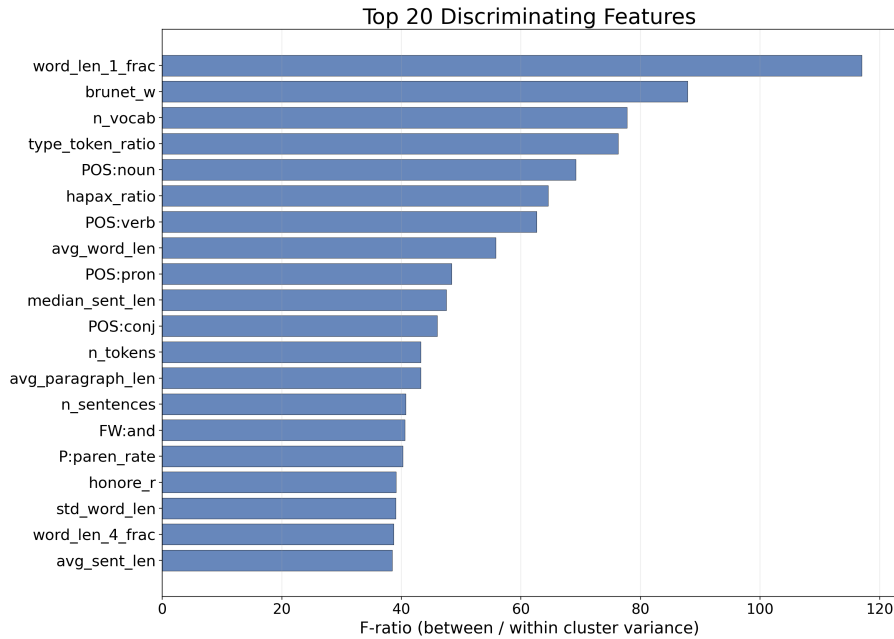


Figure 4: Top 20 discriminating features for the $K = 10$ clustering, ranked by ANOVA F-ratio. Single-character word rate (`word_len_1_frac`), Brunét’s W , vocabulary size (`n_vocab`), and POS distributions dominate.

The ranking is reassuringly stylistic rather than topical: the strongest signals are how often an author uses one-letter words (mostly “I” and “a”), vocabulary richness, the noun/verb/pronoun mix, and average word length. None of these depend on what the article is *about*.

2 Effect of Articles per Author (M)

How does data density, represented by the number of articles per author M , affect performance? An author’s writing style varies between articles, often substantially: they may write across multiple topics and genres, each dragging the surface features in different directions. An opinion piece tends to use the word “I” more often than a report, for example, and the rate of “I” is one of our features. LessWrong articles also carry residual noise from code blocks, MathJax formulas, image captions, quoted material, and audio transcripts that are difficult to remove perfectly. With too few articles per author, the sample becomes a poor estimate of that author’s true style, so we expect K-means performance to degrade as M shrinks.

Setup. We reuse the LessWrong 10 Authors corpus, fix $K = 4$ and length $N = 1,526$ words, and sweep $M \in \{5, 10, 15, 20, 25, 30\}$. For each M we run 50 trials: sample 4 of the 10 authors at random, sample M articles per author, truncate, extract features, cluster, record ARI.

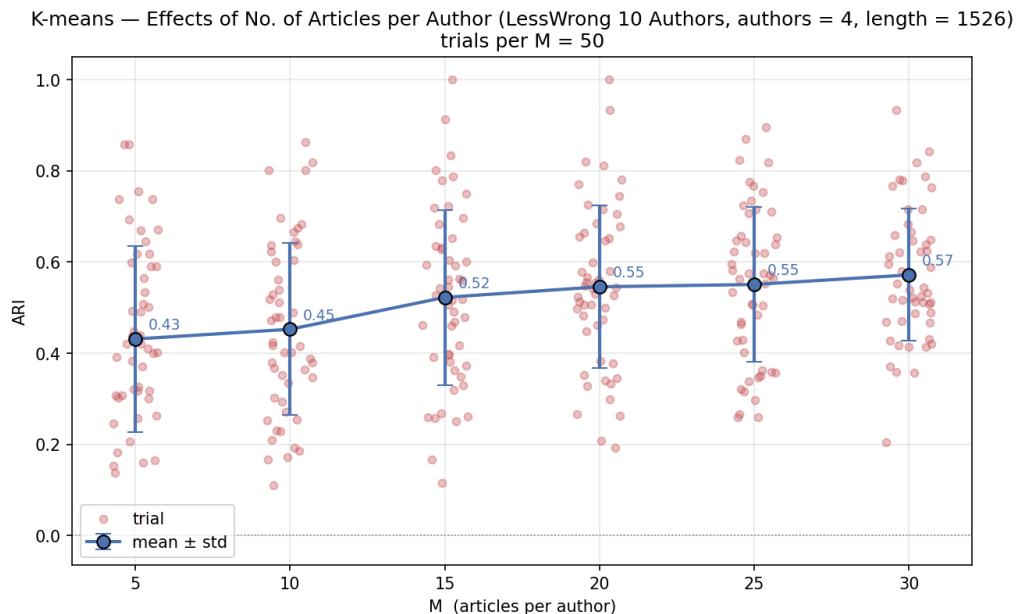


Figure 5: ARI vs. M on LessWrong 10 Authors ($K = 4$, $N = 1526$, 50 trials per M). Mean ARI rises gently from 0.43 at $M = 5$ to 0.57 at $M = 30$.

ARI trends upward from 0.43 at $M = 5$ to 0.57 at $M = 30$, but the slope is shallow and the curve appears to be levelling off. It is unclear whether the trend would continue past $M = 30$ or whether performance has effectively saturated.

To push M higher without generating many highly correlated trials (which would happen if we kept resampling similar 4-author combinations from only 10 authors), we need authors with substantially more articles. Finding 10 LessWrong authors with more than 30 qualifying articles (length $\geq 1,526$ words) is difficult. We therefore assemble a separate *LessWrong 4 Authors* dataset: four authors for whom we collected 76–79 long articles each. Conclusions will be specific to this group, but they can hint at whether chasing higher M is worth the effort.

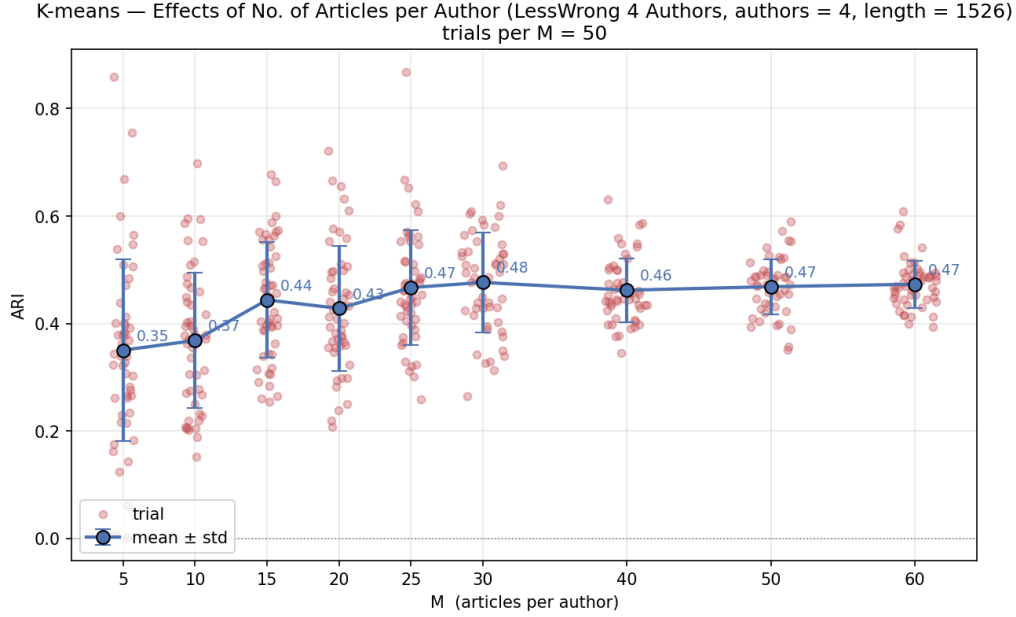


Figure 6: ARI vs. M on LessWrong 4 Authors ($K = 4$, $N = 1526$, 50 trials per M). The curve climbs from 0.35 at $M = 5$ to ~ 0.47 near $M = 25$, then plateaus through $M = 60$.

The pattern closely mirrors the 10-author case: mean ARI climbs from 0.35 to ~ 0.47 as M goes from 5 to 25, then **flattens** between $M = 25$ and $M = 60$, hovering at 0.46–0.48. The break point sits around $M = 25$ –30. On this kind of LessWrong corpus, collecting more than roughly 30 articles per author yields little extra clustering quality.

3 Effect of Number of Authors (K)

Does clustering more authors hurt performance? Intuitively yes: the more authors in the pool, the more likely some pairs of writing styles overlap, making K-means’ task harder. To isolate the effect of K from confounders, we must hold everything else fixed.

Setup. We reuse LessWrong 10 Authors and fix $M = 10$ articles per author and $N = 1,526$ words. For each $K \in \{2, 4, 6, 8, 10\}$ we run 50 trials: sample K of the 10 authors at random, sample $M = 10$ of each chosen author’s 40 articles, truncate, extract features, cluster, record ARI.

A note on $K = 10$: there is exactly one way to choose all 10 authors, so the variance at $K = 10$ comes only from article sampling and `k-means++` initialisation, not from author-group sampling. At smaller K , author-group sampling contributes an additional source of variance.

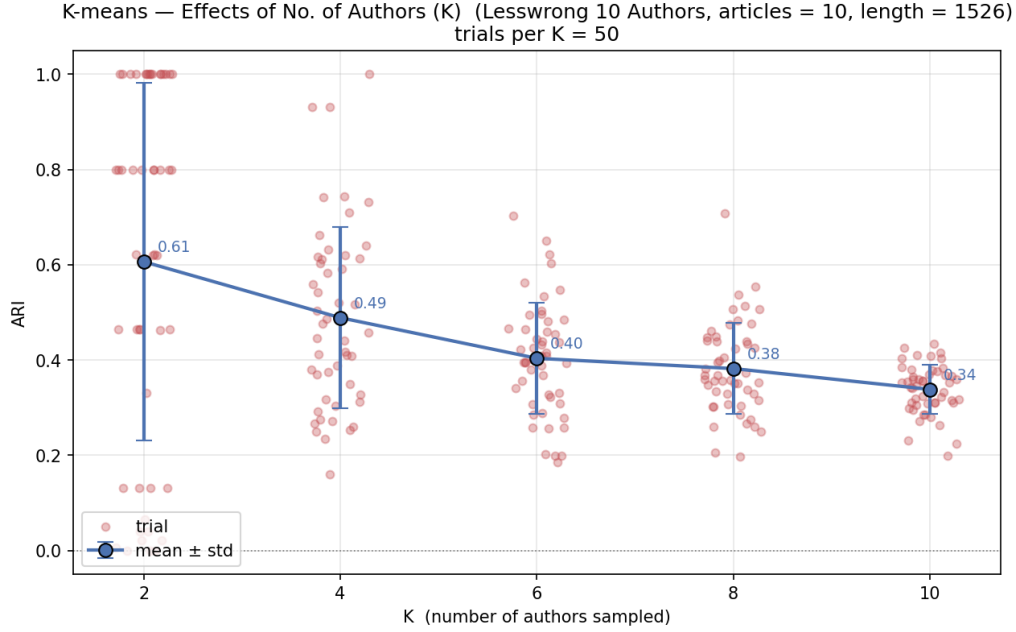


Figure 7: ARI vs. K on LessWrong 10 Authors ($M = 10$, $N = 1526$, 50 trials per K). Mean ARI drops monotonically from 0.61 at $K = 2$ to 0.34 at $K = 10$.

The downward trend is clear. Performance degrades steadily as K grows, in line with the intuition that more authors means more style overlaps.

4 Effect of Article Length (N)

How does article length N affect performance? Most of our 107 features are *rates* (proportions, frequencies, means): quantities that should be agnostic to length in expectation, yet noisier when estimated from short passages.

Setup. We use LessWrong 10 Authors with $K = 5$ and $M = 15$, sweeping $N \in \{100, 200, 500, 1000, 1500\}$. For each N we run 50 trials: sample 5 of the 10 authors, sample 15 articles per author, truncate to the first N words, extract features, run K-means.

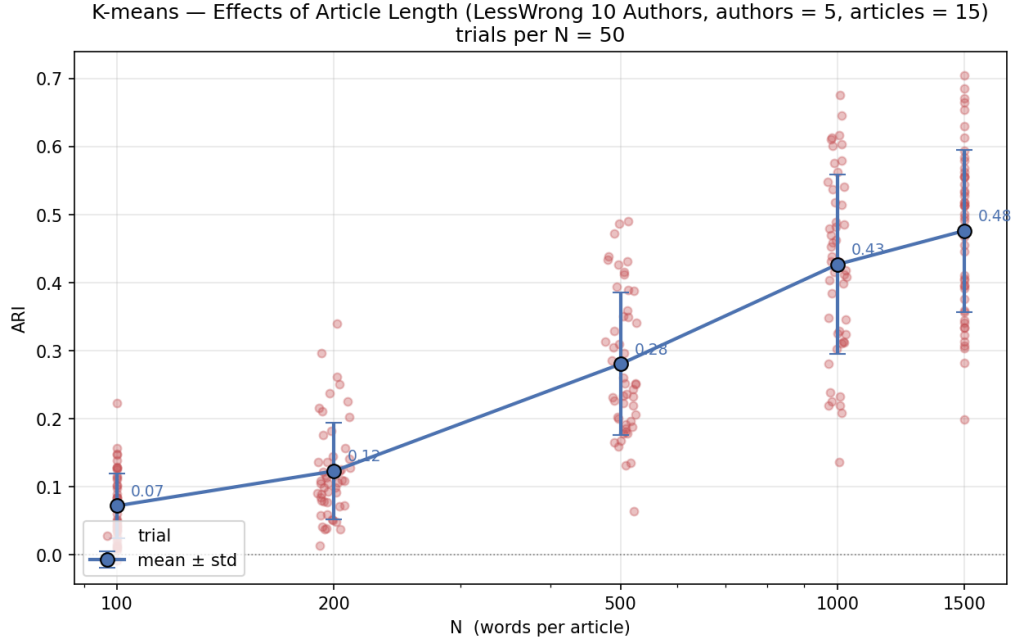


Figure 8: ARI vs. N on LessWrong 10 Authors ($K = 5$, $M = 15$, 50 trials per N). ARI rises steeply from 0.07 at $N = 100$ to 0.48 at $N = 1500$.

The trend is steep: ARI climbs from 0.07 at $N = 100$ to 0.48 at $N = 1500$, with the slope easing slightly between $N = 1000$ and $N = 1500$, hinting at a plateau that has not yet arrived. To see whether the curve really flattens, we need longer articles. Finding many long (≥ 3000 -word) articles for ten LessWrong authors is impractical, so we switch to a *LessWrong 5 Authors* dataset: five authors for whom we found 35–57 articles each of length $\geq 3,000$ words. The setup is identical except there is no author sampling (all 5 are used each trial) and N now ranges up to 3,000.

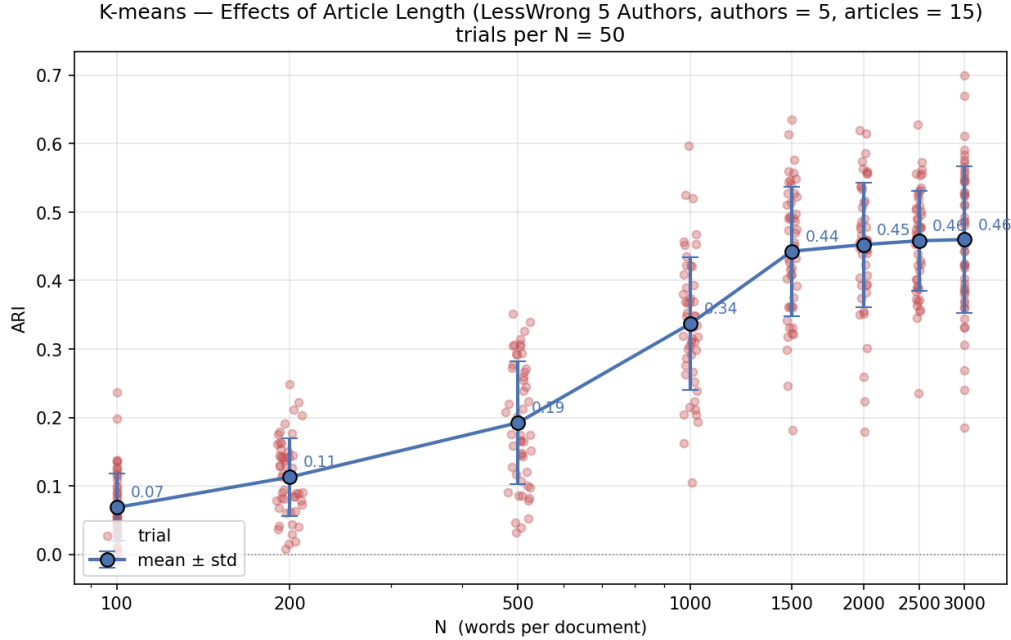


Figure 9: ARI vs. N on LessWrong 5 Authors ($K = 5$, $M = 15$, 50 trials per N). The curve follows the 10-author trend up to $N = 1500$, then **flattens**: ARI moves only from 0.44 to 0.46 over the range $N \in \{1500, 2000, 2500, 3000\}$.

The shape is striking. Up to $N = 1500$ the curve mirrors the 10-author result; past $N = 1500$ it hits a clear break point, barely moving from 0.44 to 0.46 over the next 1,500 words. A few possible explanations:

- **Feature convergence.** By $N = 1500$ most rate-based features may already be close to their author means, so the algorithm is hitting a performance ceiling for this dataset and feature set.
- **Topic shifts within long articles.** Long articles often span multiple sub-discussions; transitions between them can shift the stylistic surface. The break point may reflect an implicit average coherent-stretch length of roughly 1,500 words.
- **Noisy tails.** Authors often write discursively at the start and then move into code blocks, image captions, math, and quoted material, all of which add noise rather than signal to the features.
- **Dataset-specific.** The break point may simply be a property of this particular five-author group rather than a general pattern.

Regardless, the practical implication is the same: there is little to gain from chasing articles longer than $\sim 1,500$ words for this kind of clustering problem.

Why does the curve rise in the first place? Of our 107 features, roughly 100 are proportions, rates, means, or medians that should, in expectation, reflect the author’s intrinsic style independent of article length. But by the Law of Large Numbers, these quantities only converge to their author-specific means with a large sample. At small N each feature is a noisy estimate of the underlying rate, causing articles from different authors to overlap in feature space.

To make this concrete, we isolate author `zvi` from LessWrong 5 Authors and track two features as a function of passage length:

1. Concatenate all of `zvi`’s articles into a single “master” document.

- For each window size $N \in \{100, 200, 500, 1000, 1500, 2000, 2500, 3000\}$, draw 20 random contiguous slices of exactly N tokens.
- Compute `fw_you` (rate of the word “you”) and `word_len_1_frac` (rate of one-character words) on each slice.

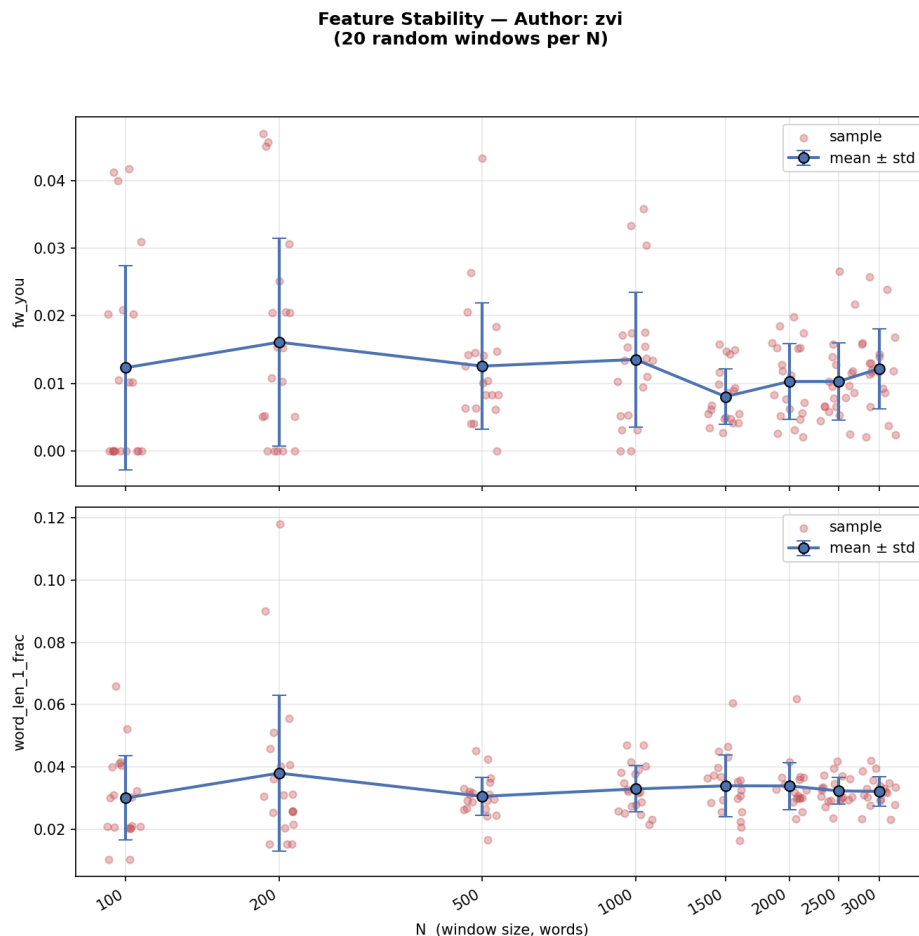


Figure 10: Two stylometric features computed on random windows of `zvi`’s concatenated text. The spread around each mean narrows as N grows.

The standard deviation of both features is large at small N and shrinks roughly as $1/\sqrt{N}$, the textbook behaviour of a sample mean. At $N = 100$ each window is too short to yield a reliable estimate of `zvi`’s rates; by $N = 1500$ the estimates have stabilised. This mechanism drives the rising portion of the ARI-vs- N curve.

5 Effect of Era

Why did we collect equal numbers of articles from ≤ 2022 and ≥ 2024 for the LessWrong 10 Authors dataset? A practical observation motivated it. ChatGPT and similar chatbots emerged in late 2022 and quickly gained traction as writing assistants. By 2024 they were routine tools for many writers, used to proofread or polish blog posts, emails, and similar texts. Chatbots have a distinctive style of their own; if a substantial fraction of authors run their drafts through one, individual writing styles might homogenise. We therefore hypothesised **lower K-means performance in the post-2024 era than in the pre-2022 era**.

Setup. LessWrong 10 Authors, $K = 4$, $M = 15$, $N = 1,526$, 50 trials per era. Each trial samples 4 of the 10 authors and 15 of each author’s 20 era-restricted articles; the rest of the pipeline is unchanged. The two eras are sampled independently, making their error bars statistically comparable.



Figure 11: ARI by era ($K = 4$, $M = 15$, $N = 1526$, 50 trials per era). Mean ARI drops from **0.71** in the pre-2022 era to **0.56** in the post-2024 era.

The mean ARI drops from **0.71** (pre, ≤ 2022) to **0.56** (post, ≥ 2024), a substantial gap that is consistent with the AI hypothesis.

Does this prove people are using AI to polish their writing? Unfortunately, no. The drop is too large to dismiss as noise, but AI use is only one of several plausible explanations:

- **Writing evolves.** Style is not static. An author’s prose today may not mirror their prose from five years ago simply because they have grown intellectually and socially.
- **Topic shifts on LessWrong.** The community’s focus has changed in response to current events. For instance, `kaj_sotala`’s pre-2023 articles are mostly about biology, social dynamics, and self-improvement, while his post-2023 articles are mostly about AI. A shift in topic mix can drag surface stylistic features along with it, even when the author’s underlying voice is unchanged.

The era effect is real and worth noting; attributing it specifically to AI-assisted writing would require a controlled study that this experiment cannot deliver.